PAGE 1 - Atoms

An atom has a nucleus of protons and neutrons, with electrons arranged in shells around it. The atomic number is the number of protons, and it identifies the element. Isotopes of the same element have the same number of protons but different numbers of neutrons, so their masses differ.

PAGE 2 - The periodic table

The periodic table orders elements by atomic number. Elements in the same group have the same number of outer-shell electrons, so they react in similar ways. The noble gases in group 18 have full outer shells and are very unreactive. The alkali metals in group 1 react vigorously with water.

PAGE 3 - Chemical bonds

An ionic bond forms when a metal atom transfers electrons to a non-metal atom, as in sodium chloride. A covalent bond forms when two non-metal atoms share a pair of electrons, as in water. In metallic bonding, positive ions sit in a sea of delocalised electrons.

PAGE 4 - Moles

One mole of any substance contains 6.02 x 10^23 particles, the Avogadro constant. Molar mass is the mass of one mole in grams per mole. The number of moles equals the mass divided by the molar mass, so 18 grams of water is one mole.

PAGE 5 - Reaction rates

Collision theory says particles react only when they collide with enough energy. The rate increases at a higher temperature, a higher concentration or a larger surface area. A catalyst speeds up a reaction by providing a pathway with a lower activation energy and is not used up.

PAGE 6 - Acids and bases

The pH scale runs from 0 to 14. Acids have a pH below 7 and release hydrogen ions in water; bases have a pH above 7. Each step of one pH unit is a tenfold change in hydrogen ion concentration. In neutralisation an acid and a base react to form a salt and water.

PAGE 7 - Redox

Oxidation is the loss of electrons and reduction is the gain of electrons; in a redox reaction both happen together. The rusting of iron is an oxidation: iron reacts with oxygen and water to form hydrated iron oxide.

PAGE 8 - Equilibrium

In a closed system a reversible reaction reaches dynamic equilibrium, where the forward and reverse reactions run at the same rate and the amounts stop changing. If conditions change, the position of equilibrium shifts to oppose the change. Raising the pressure favours the side with fewer gas molecules.

PAGE 9 - Revision guide

Before the chemistry quiz on Wednesday, practise mole calculations, balance redox equations, and explain how a catalyst changes the rate of a reaction. The quiz has 20 questions.

PAGE 10 - Teacher update

Update to the revision guide: the chemistry quiz moved from Wednesday to Thursday, and it now has 25 questions instead of 20.